Quotation No : Q26000244

Date : 18-May-2026

Sharp Tools Pvt. Ltd.
15, Industrial Area
PAONTA SAHIB
HIMACHAL PRADESH
173205

Kind Attention : Mr./Ms. Manish Puri

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Arifa Zaidi
Assistant Manager Business Development
9218108184
www.absstem.com



nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN UH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Product Description

a. Product Name	Ultra Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN UH**
c. Purity Range	99.99% - 99.999%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generation System
- N_2 Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA System is stored in this for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

- Guaranteed assurance of plant performance.
- Genuine spare parts and consumables being used to provide longevity and durability.
- Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
- 24 x 7 On-Call service support to our customers for any basic trouble shooting
- Emergency visit to attend any unplanned breakdown to any location within 48 hours.
- No additional surprise charges as all breakdown spares and consumables covered.
- The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

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Product Description

Product	Description	Make	Qty
Compressed Air System	C007F0046 Air System		1

Caption	Specification
Air Compressor KW	7.5 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	500 Liters Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	46 CFM
Air Pressure	7.5 Barg
POWER SUPPLY	400 V 3 Phase 50 Hz

Product	Description	Make	Qty
nGen Nitrogen Generator Ultra Purity	nGEN007UH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99.99% PURITY	14 Nm3/Hr DRY AIR REQUIRED 40 CFM
FLOW AT 99.995% PURITY	13 Nm3/Hr DRY AIR REQUIRED 45 CFM
FLOW AT 99.999% PURITY	10 Nm3/Hr DRY AIR REQUIRED 47 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	500 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	500 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*

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Product	Description	Make	Qty
Gas Storage Vessel	AI-500L Storage Vessel		1

Caption	Specification
Water Volume	500 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	3.25 M3
Corrosion Allowance	1.5 MM

Product	Description	Make	Qty
LP Booster 20 Barg	MOC 10 LP 20 Barg Booster	ABSSTEM	1

Caption	Specification
Gas to be compressed	Oxygen/Nitrogen
Suction Pressure	4 to 5 Barg
Discharge pressure	15 to 20 Barg
Voltage	415 V AC
Gas Flow	10 NM3/hr
Type	Air Cooled Oil Free Reciprocating Booster
Input Power	3 KW

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	ACSC007F	C007F0046	Compressed Air System	840510	1	6,40,000.00	18.00	6,40,000.00
2	nGEN007UH	nGEN007UH	nGen Nitrogen Generator Ultra Purity	840510	1	13,71,000.00	18.00	13,71,000.00

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3	AIS0058ST	AI-500L	Gas Storage Vessel	731100	1	60,000.00		18.00	60,000.00
4	BC4	MOC 10 LP 20 Barg	LP Booster 20 Barg	84144030	1	5,95,000.00		18.00	5,95,000.00

Sub Total	26,66,000.00
GST(%) @18.00	4,79,880.00
Grand Total (INR)	31,45,880.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Arifa Zaidi

Assistant Manager Business Development

9218108184

www.absstem.com

PSA vs. Cylinder Supply: A Cost-Benefit Analysis

Executive Summary

This report presents a comprehensive financial analysis comparing the current expenditure on cylinder supply with the projected costs and savings of implementing an on-site PSA (Pressure Swing Adsorption) generation plant. The evaluation indicates substantial long-term financial benefits, a rapid return on investment, and enhanced operational autonomy.

Input Parameters

Gas Type:	nitrogen
Cylinders per Day:	10
Cylinder Volume:	10 m^3
Running Hours:	10 hrs/day
Running Days:	30 days/month
Cylinder Cost:	1000 ₹/m^3
Purity:	99.999%
Power Cost:	8 ₹/kWh
Investment Cost:	₹26,66,000
Annual Maintenance:	₹1,50,000

Cylinder System Costs

Gas Cost per m^3:	₹100.00
Monthly Expense:	₹3,00,000
Operator Cost (Annual):	₹0
Rental Cost (Annual):	₹60,000
Total Annual Cost:	**₹36,60,000**

PSA System Costs

Gas Cost per m^3:	₹8.22
Power Consumption:	10.50 kW
Operator Cost (Annual):	₹0
Utilization Factor:	0.98
Total Annual Cost:	**₹4,22,160**

.td.
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Financial Summary & Investment Analysis

₹2,75,348
Estimated Monthly Savings

₹32,14,174
Estimated Annual Savings

120.6%
Return on Investment (ROI)
Payback in 10.0 months

Visual Cost Comparison

Monthly & Annual Cost Comparison



Return on Investment (ROI)



Ltd.
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Business Case & Recommendation

Transitioning from cylinder supply to an on-site PSA plant presents a compelling financial and operational advantage. With projected monthly savings of ₹2,75,348 and annual savings of ₹32,14,174, the initial investment is quickly recovered, leading to significant long-term cost reduction.

The calculated return on investment of 120.6%, with a payback period of just 10.0 months, underscores the financial viability of this project. Beyond the numbers, on-site generation eliminates dependence on external suppliers, mitigates logistical risks, and reduces the carbon footprint associated with cylinder deliveries.

Recommendation: We strongly recommend proceeding with the implementation of the PSA generation system to realize immediate cost savings, improve operational efficiency, and achieve supply chain independence.

10-Year Cumulative Savings

YEAR	CURRENT SYSTEM COST	PSA SYSTEM COST	CUMULATIVE SAVINGS
Year 1	₹36,60,000	₹4,22,160	₹5,71,840
Year 2	₹73,20,000	₹8,44,320	₹38,09,680
Year 3	₹1,09,80,000	₹12,66,480	₹70,47,520
Year 4	₹1,46,40,000	₹16,88,640	₹1,02,85,360
Year 5	₹1,83,00,000	₹21,10,800	₹1,35,23,200
Year 6	₹2,19,60,000	₹25,32,960	₹1,67,61,040
Year 7	₹2,56,20,000	₹29,55,120	₹1,99,98,880
Year 8	₹2,92,80,000	₹33,77,280	₹2,32,36,720
Year 9	₹3,29,40,000	₹37,99,440	₹2,64,74,560
Year 10	₹3,66,00,000	₹42,21,600	₹2,97,12,400

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